ceramic electrochemical cells (PCECs), enhancing their performance below 450°C when combined with optimal electrolytes. In testing their new PCECs, they achieved high power densities in fuel-cell mode and remarkable current densities in steam electrolysis mode. These PCECs were also effective using direct methane and ammonia for power generation below 450°C and maintained stability at 400°C. Initial results suggest these PCECs hold potential for efficient fuel cells and hydrogen generation at lower temperatures, with the developed electrolyte offering resistances on par with or better than previous PCECs crafted through complex methods.

New OLED with ultra-low voltage blue emission

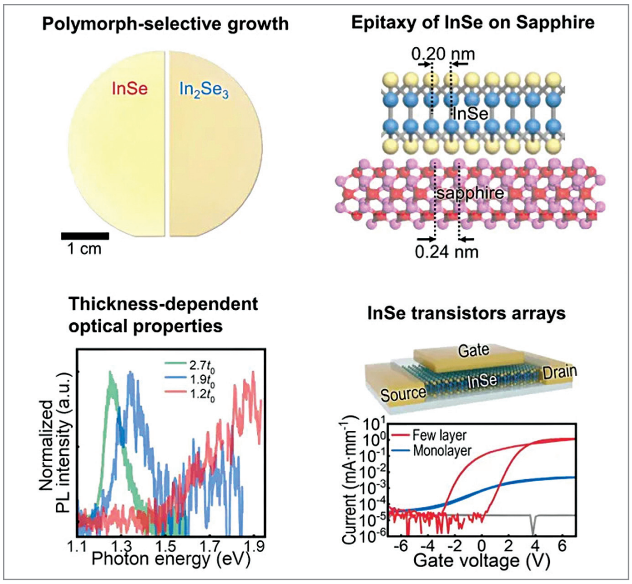
In collaboration with the University of Toyama, Shizuoka University, and the Institute for Molecular Science, researchers from the Tokyo Institute of Technology and Osaka University have showcased a unique OLED device.

Lighting up a blue organic LED with a single AA battery (Credit: Associate Professor Izawa and the member and authors of this research team)

This device boasts an impressively low turn-on voltage of 1.47V for blue emission, with a peak wavelength at 462nm (2.68eV). It operates using an upconversion mechanism where holes and electrons move into the donor and acceptor layers, merging at their boundary to form a charge transfer state. The innovative low-voltage blue-light-emitting OLED employs a fluorescent emitter typical in mainstream displays. This development marks significant progress towards meeting commercial blue OLED standards, emphasising the importance of refining the D/A interface to control excitonic processes, potentially benefiting OLEDs, organic photovoltaics, and various organic electronic devices.

Advanced 2D semiconductor wafers for modern chip integration

The University of Pennsylvania researchers have cultivated a 2D semiconductor that aligns with industrial-scale wafer dimensions. Integrating a 2D semiconductor atop its silicon counterpart could minimise chip surface area encapsulated within a delicate film. Leveraging the vertical metal-organic chemical vapour deposition (MOCVD) technique surmounted the challenges of achieving a balanced 50:50 chemical structure across a substantial area. This approach pivoted on sequentially introducing indium and selenium, thereby preventing unwanted chemical structures, and ensuring a uniform elemental ratio across the entire wafer. The team aligned the crystal directions within the material, fostering a conducive environment for electron transport, further augmenting the semiconductor's quality. This achievement meets essential industry standards and is set to transform semiconductor manufacturing, heralding enhanced purity, scalability, and performance in advanced technology.



Graphical abstract (Credit: Matter, 2023)

Revolutionising lithium-ion battery recycling

Researchers from the Chinese Academy of Sciences have unveiled a solution that leverages the electron transfer occurring during liquid-solid contact electrification to generate free radicals that initiate crucial chemical reactions. The researchers have introduced a 'mechano-catalytic' method called contact-electro-catalysis that uses radicals from contact electrification and recyclable SiO2 as a catalyst to enhance metal leaching under ultrasonic waves. The method proved efficient for recycling lithium and cobalt in exhausted lithium-ion batteries, with a 100% lithium and 92.19% cobalt leaching efficiency for lithium cobalt (III) oxide batteries and over 94% efficiency for various metals in ternary lithium batteries, all achieved within six hours. This method emerges as a pivotal solution for sustainable electronics industry recycling, with future studies set to refine its application and assess its real-world viability.